```
D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-PIKAN_Official\experiments\Volterra1D_MCPI.py
The network model is PINNs
Layers: ModuleList(
  (0): Linear(in_features=1, out_features=20, bias=True)
  (1): Linear(in_features=20, out_features=20, bias=True)
  (2): Linear(in_features=20, out_features=1, bias=True)
)
Activation function: Tanh()
Total trainable parameters: 481
Model on device: cuda:0
Start training...
Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
Training: 100%|██████████| 2000/2000 [00:05<00:00, 334.85epoch/s, Loss=1.52e-04, LR=1.0e-03, Best=1.52e-04]
Training finished. Best loss: 9.62e-05. Model saved to results/best_model.pth
Loss history saved to results/loss_history.npy
Training time: 6.788896322250366s
Relative loss: 0.011877505108714104
L2 loss: 0.007557672914117575
Parameter containing:
tensor([[-1.2629],
        [-0.1445],
        [ 0.8250],
        [-1.1281],
        [-1.7817],
        [ 0.3340],
        [-0.5699],
        [ 1.5006],
        [ 0.9744],
        [-0.5442],
        [-1.2742],
        [-0.6005],
        [ 1.5875],
        [-0.0763],
        [ 0.0408],
        [-0.5988],
        [-0.4236],
        [-0.3122],
        [ 1.3148],
        [-0.7374]], device='cuda:0', requires_grad=True)
Parameter containing:
tensor([ 1.0891, -0.1510, -0.2548,  0.8305,  1.5597, -0.4154, -0.1276,  0.2030,
        -0.1960, -0.5388, -0.6640, -0.2087, -0.4083,  0.6654,  0.9365, -0.3815,
        -0.1975, -0.4854,  0.2058, -0.6908], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([[ 3.6258e-01, -1.9885e-01,  2.5288e-02,  3.0939e-01,  3.4178e-01,
          6.9087e-02,  1.9504e-01, -1.2149e-02, -1.4326e-01,  1.3459e-01,
          3.2568e-02,  2.2975e-02, -6.1078e-02,  2.1686e-01,  1.1151e-01,
         -9.7904e-03, -1.2445e-01,  4.8888e-02, -1.2598e-01,  2.0868e-02],
        [-9.0006e-01, -1.4967e-01,  8.0751e-01, -1.2751e+00, -1.1307e+00,
          2.3355e-01, -1.9506e-01, -4.0749e-02,  4.6212e-01,  2.3114e-01,
```

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          1.5470e-01, -2.5534e-01, -1.7141e-01, -2.0357e-01, -2.0706e-01,
          2.0167e-02, -1.8990e-01, -1.3457e-01, -2.1745e-01,  1.1606e-01],
        [ 2.6437e-01, -9.9585e-02, -2.2007e-01,  3.5876e-01, -3.1667e-02,
          9.8597e-02,  3.1578e-02,  2.0540e-01, -6.6552e-02,  9.0458e-02,
          9.8375e-02,  1.1643e-01, -1.0452e-01,  5.6802e-02,  1.1462e-01,
         -1.9406e-01,  1.0096e-01,  5.8229e-02, -1.4349e-02, -2.8126e-02],
        [-5.9386e-03, -2.7453e-01,  2.3202e-02, -3.2456e-01, -2.0366e-01,
         -1.4453e-01, -1.3511e-01,  1.6275e-01,  1.6505e-01,  1.1588e-01,
          1.3948e-01, -2.0002e-01, -2.3376e-01,  4.4040e-02, -1.2865e-01,
          1.5906e-01,  1.1069e-01, -5.1657e-04,  6.4050e-02, -1.3812e-01],
        [ 1.7583e-01,  2.5400e-02, -4.8573e-01,  5.6608e-01,  4.3717e-01,
         -2.6723e-01,  2.1973e-01, -9.0541e-02, -4.7202e-01,  1.3284e-01,
          1.8670e-01,  2.4494e-01, -1.8471e-01,  1.0111e-02, -5.4362e-02,
          1.9358e-01, -8.0057e-02, -1.3541e-01,  9.6652e-02,  5.1630e-03],
        [ 8.4782e-02,  5.5132e-02, -9.5564e-02, -1.8493e-02,  9.5583e-02,
          1.3699e-02, -1.1627e-02, -8.6546e-02, -2.4614e-01,  8.1910e-02,
         -4.4805e-02, -1.5681e-01,  1.5914e-01,  8.1339e-02, -1.9541e-02,
         -9.1971e-02, -9.6803e-03,  1.6541e-01, -8.4527e-02,  1.3575e-01],
        [-7.2911e-02,  1.1983e-01, -1.3161e-01, -8.1704e-02, -3.3861e-01,
          1.9474e-01,  5.9582e-02,  1.6425e-01,  4.9473e-02,  1.9174e-01,
          2.0045e-01, -6.5509e-02,  1.4553e-01,  9.6072e-02, -1.3296e-01,
          7.2353e-02,  1.6104e-02,  8.0060e-02,  5.3468e-02, -1.1550e-02],
        [-9.2457e-02,  1.7005e-01, -8.2607e-02,  1.4001e-01,  3.0586e-01,
          2.4638e-01,  7.3853e-02, -1.7883e-03, -1.1016e-02,  2.1613e-01,
          1.3212e-01,  3.9044e-02,  2.7240e-01, -3.7503e-02,  1.4517e-01,
         -1.1045e-01,  1.1119e-02,  2.7370e-01,  2.1750e-01,  8.8639e-02],
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         -4.3195e-02, -1.1454e-02, -1.7463e-01,  2.2128e-01, -5.2437e-02,
         -2.4884e-02, -1.7514e-01, -2.1133e-02, -1.2832e-01, -1.4923e-01],
        [ 3.0916e-01,  1.0025e-01, -6.5299e-01,  2.4630e-01,  3.2595e-01,
         -9.2932e-02,  1.1082e-01, -2.8096e-04, -2.3072e-01,  1.5731e-01,
          8.8810e-02,  1.6468e-01, -3.6899e-01, -1.5238e-01, -8.4164e-02,
          1.8072e-01,  8.5468e-02,  2.1843e-01,  7.0839e-02,  5.9563e-02],
        [ 3.5599e-01,  1.3727e-01, -2.3061e-01,  7.7443e-01,  4.9568e-01,
         -3.7225e-01,  1.0872e-01,  2.3159e-01, -8.5198e-02,  5.4289e-02,
         -1.5322e-01,  7.6614e-02, -1.3818e-01,  1.0377e-01,  4.1948e-03,
          8.9839e-02,  2.0592e-01, -5.3810e-02,  9.2217e-02, -4.7751e-02],
        [ 1.8375e-02, -2.5057e-02, -1.6392e-02, -6.2967e-02, -7.4738e-02,
         -7.4002e-02, -1.0514e-01,  1.3796e-01, -2.3916e-02,  2.2517e-01,
          1.3315e-01,  7.6760e-02,  1.5022e-02,  7.4260e-02, -3.5071e-02,
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        [-9.6696e-02, -9.1235e-02, -1.5069e-01,  6.4149e-02,  9.1026e-02,
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         -6.5917e-02,  5.5915e-02,  1.4668e-01,  1.3232e-02,  1.5725e-01,
          4.8541e-02,  2.6057e-01, -7.1492e-02,  1.7536e-01, -3.9559e-02],
        [ 6.4889e-02, -2.6416e-01,  2.8178e-02,  1.0201e-02,  7.9063e-02,
         -2.1711e-01, -2.3862e-01, -1.6517e-01,  6.5622e-02, -7.5521e-03,
         -2.0666e-01, -5.0207e-02,  5.0961e-02, -1.4588e-01, -1.0860e-01,
          1.5513e-01, -1.0847e-01, -1.7184e-01, -3.1420e-02, -2.1919e-01],
        [-4.5485e-01, -3.9023e-01,  6.5562e-01, -5.3256e-01, -4.5893e-01,
          3.9628e-01, -6.0417e-01,  5.0609e-01,  8.2708e-01, -4.6555e-01,
         -2.6899e-01, -8.8817e-03,  5.6564e-01,  2.2255e-01,  3.3091e-02,
```

```
         -3.7347e-01, -2.9320e-01, -8.9979e-02,  5.2886e-01, -3.4591e-01],
        [-3.9466e-02, -7.9001e-02,  1.5492e-01, -2.1915e-01, -4.4056e-03,
          2.9430e-01, -3.2905e-02,  2.4448e-01,  3.3235e-01,  2.5705e-02,
         -9.9916e-02, -1.4687e-01,  1.9159e-01, -1.2846e-01, -1.9673e-01,
         -6.3926e-02,  1.9150e-01, -1.1997e-02, -7.8914e-03, -2.1838e-01],
        [ 9.4425e-02, -1.3243e-01, -1.8034e-01, -2.1669e-01, -8.8423e-02,
         -4.8260e-02,  1.5838e-01, -1.4895e-01,  1.2277e-02, -2.4615e-01,
         -7.7780e-02, -2.5536e-01,  4.0871e-02,  1.9231e-01,  2.6678e-01,
          1.1096e-01,  9.2647e-02, -8.3124e-03, -4.1743e-02,  2.3623e-02],
        [ 6.0189e-02, -8.8840e-02, -5.0221e-01,  4.3867e-01,  3.3166e-01,
         -2.3447e-01,  1.7044e-01, -1.5547e-01, -1.2544e-01,  1.7292e-01,
          1.7449e-02,  1.3815e-01, -2.4441e-01,  4.2429e-02,  4.4147e-02,
         -7.3224e-02,  1.7306e-01, -2.0741e-01, -9.0176e-02,  2.0135e-01],
        [-1.6529e-01,  3.0224e-02, -7.4782e-02, -1.1311e-01, -1.2340e-01,
          1.7454e-02, -1.4950e-01,  7.0777e-02,  1.4469e-01, -1.9197e-01,
          1.1524e-01,  4.8056e-02, -2.7317e-01,  3.6318e-02, -7.8176e-02,
          9.4779e-02, -2.3685e-01, -2.0850e-01, -1.3237e-01, -1.7289e-01],
        [ 2.7645e-01, -8.6685e-03, -1.2290e-02,  6.2585e-01,  1.0797e-01,
         -2.4980e-01,  1.4112e-01, -7.8208e-02, -1.9349e-01, -1.6296e-02,
         -1.8238e-01,  6.8223e-02, -2.1155e-01,  2.1790e-01,  1.7490e-01,
          2.2477e-01, -5.5245e-02, -1.6568e-01, -1.1797e-03,  5.7548e-02]],
       device='cuda:0', requires_grad=True)
Parameter containing:
tensor([ 0.2429, -0.2254, -0.0349, -0.0790, -0.1711,  0.1130,  0.1658, -0.0192,
         0.1851,  0.1804,  0.1435, -0.2208, -0.2343,  0.1864, -0.1743, -0.0571,
         0.0642, -0.1541,  0.0771,  0.2215], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([[ 0.2532, -0.1414,  0.1505, -0.1467, -0.1444,  0.0035, -0.0875,  0.2931,
         -0.1636, -0.2043,  0.4649,  0.1058,  0.0555, -0.1888,  0.4173,  0.2358,
         -0.1973, -0.1072, -0.3808,  0.1310]], device='cuda:0',
       requires_grad=True)
Parameter containing:
tensor([-0.0848], device='cuda:0', requires_grad=True)
==========================================================================================
Layer (type:depth-idx)                   Output Shape              Param #
==========================================================================================
PINNModel                                [1, 1]                    --
├─ModuleList: 1-5                         --                        (recursive)
│    └─0.weight                                                      ├─20
│    └─0.bias                                                        ├─20
│    └─1.weight                                                      ├─400
│    └─1.bias                                                        ├─20
│    └─2.weight                                                      ├─20
│    └─2.bias                                                        └─1
│    └─Linear: 2-1                         [1, 20]                   40
│    │    └─weight                                                    ├─20
│    │    └─bias                                                      └─20
├─Tanh: 1-2                               [1, 20]                   --
├─ModuleList: 1-5                         --                        (recursive)
│    └─0.weight                                                      ├─20
```

```
│       └─0.bias                                                          ├─20
│       └─1.weight                                                        ├─400
│       └─1.bias                                                          ├─20
│       └─2.weight                                                        ├─20
│       └─2.bias                                                          └─1
│       └─Linear: 2-2                        [1, 20]                     420
│       │      └─weight                                                    ├─400
│       │      └─bias                                                      └─20
├─Tanh: 1-4                                [1, 20]                     --
├─ModuleList: 1-5                          --                          (recursive)
│       └─0.weight                                                        ├─20
│       └─0.bias                                                          ├─20
│       └─1.weight                                                        ├─400
│       └─1.bias                                                          ├─20
│       └─2.weight                                                        ├─20
│       └─2.bias                                                          └─1
│       └─Linear: 2-3                        [1, 1]                      21
│       │      └─weight                                                    ├─20
│       │      └─bias                                                      └─1
==========================================================================================
Total params: 481
Trainable params: 481
Non-trainable params: 0
Total mult-adds (Units.MEGABYTES): 0.00
==========================================================================================
Input size (MB): 0.00
Forward/backward pass size (MB): 0.00
Params size (MB): 0.00
Estimated Total Size (MB): 0.00
==========================================================================================

进程已结束，退出代码为 0

```